

LAOAG, Philippines

WMO: 982230

Lat: 18.1832N

Lon: 120.5347E

Elev: 25

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	65.2	67.1	58.7	73.9	71.8	60.4	78.6	72.8	19.0	80.7	17.4	80.7	1.6	90	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.8	94.3	82.0	93.3	81.7	92.3	81.3	84.3	92.0	83.6	91.2	83.0	90.5	9.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.3	168.4	90.5	81.4	163.4	89.5	80.8	160.3	88.8	48.5	92.0	47.7	91.8	47.0	90.6	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	16.0	14.3	60.6	98.0	3.1	2.2	58.3	99.6	56.5	100.9	54.8	102.2	52.5	103.8
			58.4	85.7	2.3	0.9	56.7	86.4	55.4	86.9	54.1	87.5	52.5	88.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	77.7	78.7	81.2	83.6	84.6	83.9	83.0	82.3	82.4	82.2	81.5	79.6
	DBStd	2.94	2.69	2.49	2.29	1.79	1.97	1.91	1.91	2.09	1.73	2.01	2.08	2.53
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11582	859	805	966	1008	1073	1018	1022	1001	971	997	946	916
	CDD65	6107	394	385	501	558	608	568	557	536	521	532	496	451
	CDH74	64978	3271	3444	5305	6877	7633	6722	6067	5563	5433	5639	5024	4000
	CDH80	26107	1141	1307	2257	3207	3604	2919	2319	1918	1987	2160	1882	1406
(14) Wind	WSAvg	6.3	6.4	6.2	6.2	5.9	5.9	6.4	6.2	6.6	5.7	6.6	6.6	7.1
(15) Precipitation	PrecAvg	87.4	0.1	0.1	0.1	0.6	5.4	13.0	18.5	21.2	17.6	5.7	1.8	0.5
	PrecMax	125.7	0.9	0.7	0.9	7.9	15.9	44.7	59.3	41.9	45.4	33.9	9.4	4.7
	PrecMin	54.9	0.0	0.0	0.0	0.0	0.0	1.0	1.0	3.0	1.6	0.0	0.0	0.0
	PrecStd	19.4	0.2	0.2	0.2	1.5	4.4	9.9	14.9	10.9	10.5	6.9	2.4	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	92.3	94.1	95.2	96.0	94.7	93.4	92.5	92.7	93.6	93.5	92.2
		MCWB	76.6	76.8	78.9	81.6	82.8	83.9	83.8	83.4	82.5	80.4	79.3	77.8
	2%	DB	89.2	89.9	91.8	93.5	94.4	93.0	91.7	91.0	91.0	91.7	91.6	90.2
		MCWB	76.4	76.9	78.7	81.4	82.9	83.2	83.0	82.5	82.0	80.4	79.0	77.5
	5%	DB	87.6	88.4	90.6	92.5	93.2	91.6	90.4	89.7	89.7	90.4	90.0	88.8
		MCWB	76.2	76.8	78.5	81.1	82.4	82.6	82.2	81.9	81.4	80.2	78.8	77.1
	10%	DB	86.0	87.1	89.3	91.2	91.7	90.7	89.4	88.4	88.5	89.2	88.6	87.3
		MCWB	75.3	76.3	78.2	80.6	81.8	82.1	81.6	81.3	81.0	79.9	78.6	76.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	80.8	82.3	83.8	85.1	85.2	84.9	84.4	84.1	83.7	82.7	81.1
		MCDB	87.6	88.6	90.8	93.0	94.0	92.9	91.8	90.9	90.8	90.7	90.0	88.7
	2%	WB	78.8	79.3	81.0	82.8	83.8	84.1	83.7	83.3	82.9	82.5	81.4	79.8
		MCDB	86.5	87.3	89.6	92.0	92.6	91.7	90.6	89.6	89.4	89.5	88.8	87.4
	5%	WB	77.6	78.3	80.0	81.9	83.1	83.3	82.8	82.4	82.1	81.5	80.4	78.7
		MCDB	85.3	86.4	88.5	90.9	91.8	90.8	89.5	88.7	88.5	88.4	87.7	86.2
	10%	WB	76.4	77.2	79.1	81.1	82.3	82.5	82.0	81.5	81.3	80.5	79.5	77.4
		MCDB	83.9	85.3	87.3	89.8	90.8	89.8	88.3	87.3	87.4	87.2	86.4	84.8
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	17.3	17.0	16.2	14.8	13.5	12.5	11.5	12.8	14.1	14.7	15.5
		MCDBR	19.4	19.2	18.5	17.5	16.6	15.1	14.5	14.0	14.6	16.5	17.5	18.3
		MCWBR	9.7	9.5	9.2	8.3	7.9	7.4	7.3	7.0	7.3	7.9	8.2	8.8
	5% WB	MCDBR	17.5	17.5	16.9	16.5	15.9	14.8	14.1	13.2	14.0	14.9	15.5	16.8
		MCWBR	9.8	9.7	9.1	8.3	8.0	7.6	7.4	7.0	7.3	7.7	8.1	9.1
(40) Clear-Sky Solar Irradiance	taub		0.407	0.415	0.480	0.501	0.462	0.459	0.459	0.465	0.462	0.457	0.438	0.435
	taud		2.433	2.402	2.222	2.182	2.344	2.366	2.362	2.334	2.333	2.326	2.390	2.366
	Ebn at Noon		278	281	266	260	267	265	266	266	267	265	266	264
	Edh at Noon		35	37	46	48	41	39	40	41	41	40	36	36
(44) All-Sky Solar Radiation	RadAvg		1432	1676	1918	2089	1895	1830	1681	1533	1595	1560	1405	1272
	RadStd		69	111	89	100	137	182	183	172	142	118	125	89

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	-0.70	-0.59	-0.45	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	-0.59	-0.43	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page